Problem set 4

Olof de Blanche

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Question 1

1.1) Let $\mathbf{D} = (D_1, ..., D_J)$ be the fiscal transfer treatment vector. Let $\mathbf{W} = (W_{11}, ..., W_{N_1 1}, W_{12}..., W_{N_J J})$ be the cash transfer treatment vector. For SUTVA to be satisfied we need:

$$\begin{aligned} Y_{ij}(\mathbf{D}, \mathbf{W}) &= Y_{ij}(D_j, W_{ij}) \\ (D_j, W_{ij}) &\in \{0,1\}^2 \qquad\qquad \forall i, j \end{aligned}$$

I.e. that the observed outcome of the unit **only** depends on the value of D_j for the units group, and the values of W_{ij} for the unit. Additionally, the only possible treatment statuses of both variables is 0 or 1.

In this context the SUTVA assumption is a bit shaky. There is nothing stopping an individual to share its cash transfers with others.

1.2) We have:

$$\begin{aligned} Y_{ij} = Y_{ij}(0,0) + (D_j - D_j W_{ij})(Y_{ij}(1,0) - Y_{ij}(0,0)) + (W_{ij} - D_j W_{ij})(Y_{ij}(0,1) - Y_{ij}(0,0)) + \\ D_j W_{ij}(Y_{ij}(1,1) - Y_{ij}(0,0)) \end{aligned}$$

We get that $\tau_d = E[Y_{ij}(1,0) - Y_{ij}(0,0)]$, $\tau_2 = E[Y_{ij}(0,1) - Y_{ij}(0,0)]$ and $\tau_{dw} = E[Y_{ij}(1,1) - Y_{ij}(0,0)]$. I.e.the average individual effect of only getting treated in d-dimension, only getting treated in w-dimension and getting treated in both respectively.

For identification we have:

$$\begin{aligned} \tau_d = E[Y_{ij}(1,0) - Y_{ij}(0,0)] = E[Y_{ij}(1,0)|D_j = 1, W_{ij} = 0] - E[Y_{ij}(0,0)|D_j = 0, W_{ij} = 0] = \\ E[Y_{ij}|D_j = 1, W_{ij} = 0] - E[Y_{ij}|D_j = 0, W_{ij} = 0] \end{aligned}$$

Where the last expression have population-correspondences we are able to estimate. The important assumtpion here is (let $\mathbf{Y}_{ij}$ indicate $(Y_{ij}(0,0), Y_{ij}(1,0), Y_{ij}(0,1), Y_{ij}(1,1))$):

$$\mathbf{Y}_{ij} \perp \{D_j, W_{ij}\}$$

Which follows from the randomization of the experimnet and is used for the second equality. Relying on this assumption will show that both τ_w, τ_{dw} also are identified.

1.3) Doing the same as above the population estimators become:

$$\begin{aligned}
\tau_d &= E[Y_{ij}|D_j = 1, W_{ij} = 0] - E[Y_{ij}|D_j = 0, W_{ij} = 0] \\
\tau_w &= E[Y_{ij}|D_j = 0, W_{ij} = 1] - E[Y_{ij}|D_j = 0, W_{ij} = 0] \\
\tau_{dw} &= E[Y_{ij}|D_j = 1, W_{ij} = 1] - E[Y_{ij}|D_j = 0, W_{ij} = 0]
\end{aligned}$$

This corresponds to the population regression (with $E[Y_{ij}|D_j = 0, W_{ij} = 0] = \mu$):

$$Y_{ij} = \mu + \tau_d D_j(1 - W_{ij}) + \tau_w W_{ij}(1 - D_j) + \tau_{dw} D_j W_{ij} + \epsilon_{ij}$$

1.4) Let $\hat{\alpha}_j$ indicate estimated fixed effects of village j. If we want to test whether or not the fiscal transfer matters for consumption using a framework of random effects we set up model:

$$\alpha_j \sim N(\gamma_t, \sigma^2)$$

With $t \in \{0, 1\}$ indicating if you are treated or not. We test:

$$\begin{aligned}
H_0 :& \gamma_0 \geq \gamma_1 \\
H_1 :& \gamma_0 < \gamma_1
\end{aligned}$$

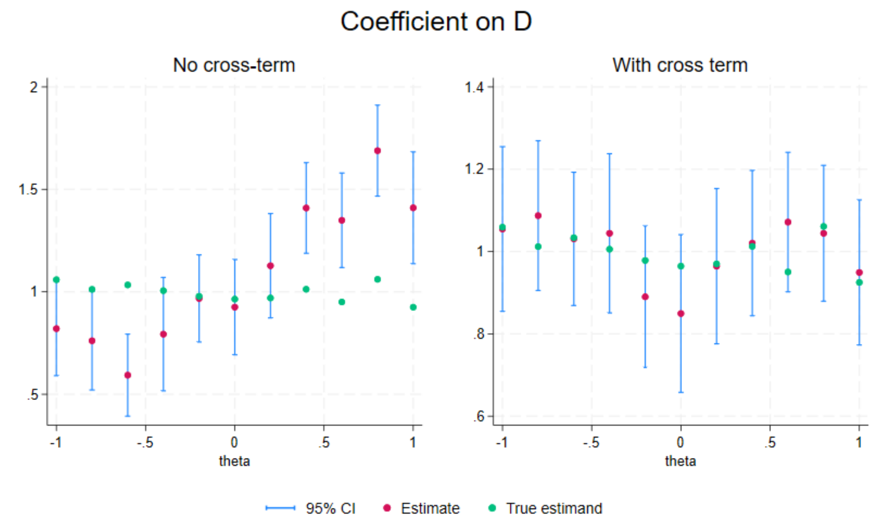
We estimate both means with $\hat{\gamma}_t = \sum_{j:D_j=t} \hat{\alpha}_j$ then perform a standard t-test.

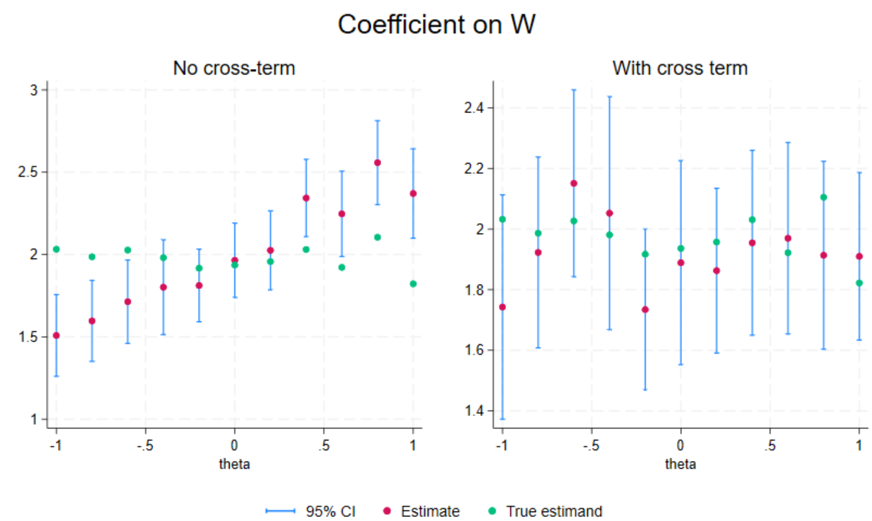
Using a fixed effects model instead, we know that if village allocation do not matter for consumption then we should have $\alpha_j = \alpha_{j'} = 0, \forall i, i'$. This can be tested using a standard F-test.

1.5) X_{ij} is a variable governing how the two treatments interact. Without it the effect of being treated in both dimensions would simply be the effect of the fiscal and cash transfer summed. ϕ governs wether the effect of being treated in both dimensions on average is smaller or larger than this sum. A negative ϕ corresponds to decreasing returns to treatment, while a positive ϕ corresponds to the opposite.

μ_j is the expected effect of a cash transfer in village j. It varies between villages.

1.6) The error in the above population specification depends on the value of treatments. This is since $Y_{ij}(1, 1)$ has a larger variance than the other PO:s. Further the error will be correlated within villages since the mean effect of cash treatment is different between villages. Hence, I use cluster robust se:s. I get the following plots:





While not controlling for the interaction term both the coefficient on D and W picks up some of the added effect coming from X_{ij}. This effect becomes larger with ϕ. When controlling for the interaction, changing ϕ has no effect on the coefficient anymore. The confidence intervals always contain the true estimand.

1.7) The coefficient on W is not affected by running it together with a dummy D or fixed effects for each village. Changing ϕ does not change this. This is by construction and comes from the fact that I have stratified over villages when constructing my dataset.

I run two different regressions:

$$\begin{aligned} &Y = \mu + \alpha D + W\beta + e \qquad\qquad \text{or} \\ &Y = F\alpha + W\beta + e \end{aligned}$$

The model is stacked and the D vector is hence on a individual level and not village level as it is described above. F is a matrix of indicator functions, $(F)_{ib} = F_{ib} = \mathbf{1}(j = b)$ By FWL:

$$\hat{\beta} = ((M_D W)' M_D W)^{-1} (M_D W Y)$$

Since we have W_{ij} is 1 a half of the time both within each group and within any value of D it follows that in the first regression we have:

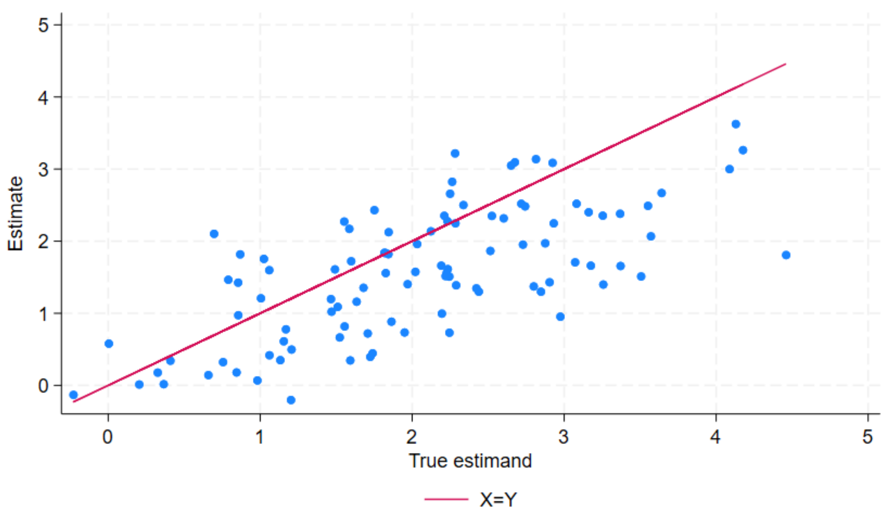
$$M_D W = W - \hat{E}[W|D] = W - \frac{1}{2} * \mathbf{1}$$

In the second regression we have:

$$M_D W = W - \hat{E}[W|F] = W - \frac{1}{2} * \mathbf{1}$$

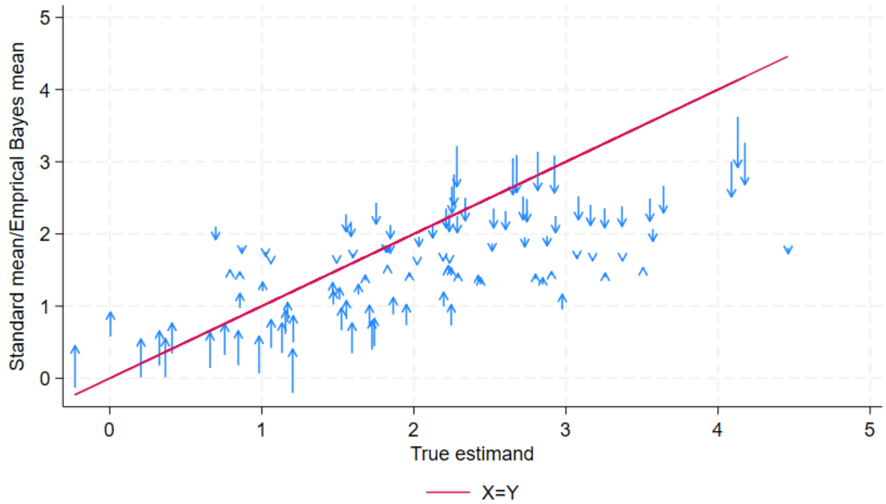
Since conditioning on F essentially the same as conditioning on being in a specific group. Hence $\hat{\beta}$ is the same in both regressions above. If there was some variance in the sample expectation of W within any of these groups we would not get the same coefficient.

1.8) I get the plot:



Where the coefficients come from a regression without a constant. The estimates are a bit inconsistent because of the small groups.

1.9) Using empirical Bayes the estimates changes as:



The vast majority of estimates are getting closer to the $X = Y$ line. At the same time it is hard to see if it results in more or less bias.

Question 2

2.1) To estimate these parameters we will run the stacked regression:

$$Y = X\theta + e$$

With the sample matrix X as (x_i:s are column vectors):

$$X = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Since its column space is 4-dimensional we know that the 6 column vectors must be linearly dependent. We for example have $x_1 + x_2 = x_3 + x_4$. We necessarily can only have 4 linearly independent columns in this case. If we drop α_1 and γ_1, the column space is only four dimensional and results in the sample matrix:

$$X = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

Which has a non-zero determinant and hence is rank 4 and invertible. We can estimate the vector $\theta = (\alpha_2, \gamma_2, \psi_1, \psi_2)$. But α_2, γ_2 now only identifies the difference between individual two and one, or time period 2 and 1.

2.2) θ is not identified. To estimate $\gamma_2 - \gamma_1$ (normalize t= 1) we would need to be able to estimate:

$$E[Y_{it}|\min\{\alpha_i, \psi_{j(i,t)} = \psi_{j(i,t)}\}] - E[Y_{it-1}|\min\{\alpha_i, \psi_{j(i,t-1)} = \psi_{j(i,t)}\}]$$

But in this setup the last term might miss a sample correspondence since the machine is changed between periods. Because of this, if $\exists i : \psi_1 < \alpha_i < \psi_2$ the time fixed effects are not identifiable.

If we want to estimate $\alpha_2 - \alpha_1$ we would need to estimate:

$$E[Y_{1t}|\min\{\alpha_1, \psi_{j(1,t)}\} = \alpha_1] - E[Y_{2t}|\min\{\alpha_2, \psi_{j(2,t)}\} = \alpha_2]$$

Which is obviously not identifiable if $\alpha_1, \alpha_2 > \psi_1, \psi_2$. The same goes for the machine fixed effects.

2.3) Adding another worker in the first case makes the θ still not identifiable. The dimension of the column space is now 6, but we will have another parameter α_3 to estimate so there will be 7 column vectors. Dropping α_3 we would still have to drop a time FE as well since the resulting sample matrix will still have perfectly collinear column vectors.

The second example is not solve either, the time fixed effects can now be identified (since we hav a worker at the same machine at two time periods), but if we still have $\alpha_1, \alpha_2 > \psi_1, \psi_2$ no individual fixed effects can be identified.

2.4) With the model:

$$Y_{it} = \alpha_i + \gamma_t + \psi_{j(i,t)} + \epsilon_{it}$$

We have:

$$\begin{aligned} E_i[Y_{i2}] - E_i[Y_{i1}] &= \gamma_2 - \gamma_1 + E_i[\psi_{j(i,2)}] - E_i[\psi_{j(i,1)}] = \\ &= \gamma_2 - \gamma_1 \end{aligned}$$

$E_i[\psi_{j(i,2)}] = E_i[\psi_{j(i,1)}]$ comes from the fact that both workers are equally likely to be assigned to both machines in our sample. Assuming $\gamma_1 = 0$ (having $t = 1$ as reference) means that:

$$\hat{\gamma}_2 = \hat{E}_i[Y_{i2}] - \hat{E}_i[Y_{i1}] = 3.5 - 2.5 = 1$$

Further:

$$\begin{aligned} E_{it}[Y_{it}|j(i,t) = 2] - E_{it}[Y_{it}|j(i,t) = 1] &= (E_i[\alpha_i] + E_t[\gamma_t] + \psi_2) - (E_i[\alpha_i] + E_t[\gamma_t] + \psi_1) = \\ &= \psi_2 - \psi_1 \end{aligned}$$

With ψ_1 as reference we get:

$$\hat{\psi}_2 = \hat{E}_{it}[Y_{it}|j(i,t) = 2] - \hat{E}_{it}[Y_{it}|j(i,t) = 1] = 3.5 - 2.5 = 1$$

Finally:

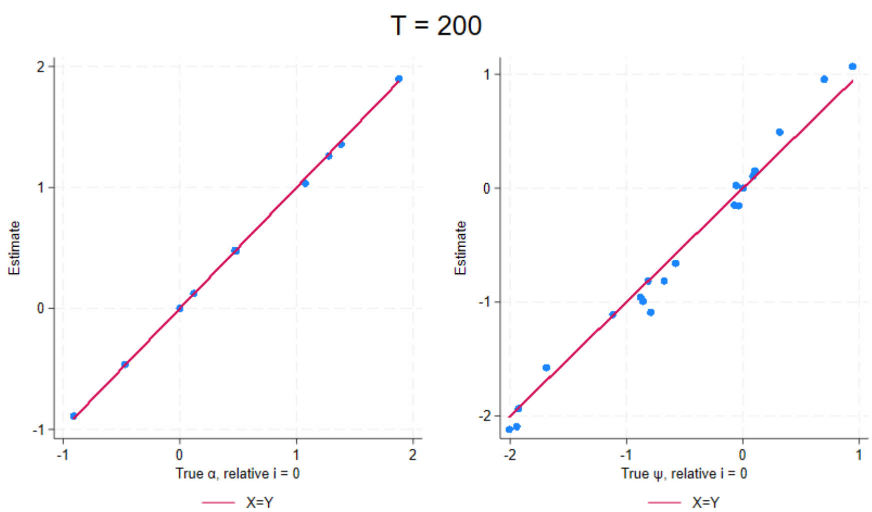
$$\begin{aligned} E_t[Y_{it}] &= \alpha_i + E_t[\gamma_t] - E_t[\psi_{j(i,t)}] && \iff \\ E_t[Y_{it}] - E_t[\gamma_t] - E_t[\psi_{j(i,t)}] &= \alpha_i \end{aligned}$$

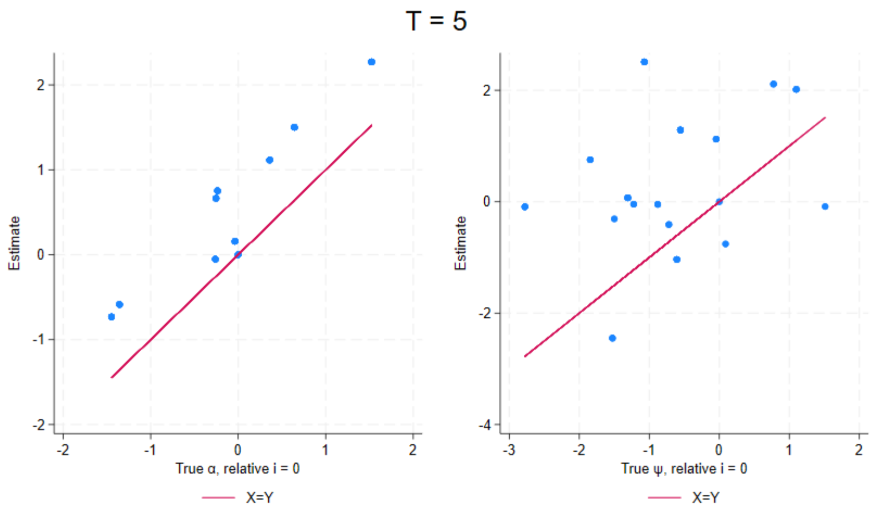
The sample correspondence become:

$$\hat{\alpha}_1 = \hat{E}_t[Y_{1t}] - \hat{E}_t[\gamma_t] - \hat{E}_t[\psi_{j(1,t)}] = 3 - 0.5 - 0.5 = 2$$
$$\hat{\alpha}_2 = \hat{E}_t[Y_{2t}] - \hat{E}_t[\gamma_t] - \hat{E}_t[\psi_{j(2,t)}] = 3 - 0.5 - 0.5 = 2$$

2.6) For the model to be identified we need that the probability of being assigned to a machine for a worker should be uncorrelated with the error term. I.e. if workers can chose to which machine to be assigned, and some workers know in beforehand which would be better the coming period, then the identification could break.

2.7-8) The plots become:





Here we see how the consistency of the FE:s are very dependent on the number of time periods.

2.9) In the perfect substitutes model the worker machine assignment does not matter. Hence they need not change their assignment process. If instead they realize that they are not perfect complements (indicate this as $\psi_j(i,t)$) then they should change the function to.

$$j(i,t) = \begin{cases} \arg\max_j \psi_j(i,t) \text{ if } \nexists i' : \arg\max_j \psi_j(i,t) = \arg\max_j \psi_j(i',t) = j^* \wedge \psi_{j^*}(i,t) < \psi_{j^*}(i',t) \\ \arg\max_{j \in J \setminus \{j^*\}} \psi_j(i,t) \text{ if ...(same condition exclude two machines now)} \\ \text{....recursively....} \\ \text{....} \end{cases}$$

I.e. the company pairs the worker with the machine it has the most synergy with at the moment. If there is another worker who also has the best synergy, and better synergy, with this machine then the worker has to pick the machine it has second best synergy with etc...

Since $\psi_j(i,t)$ is not constant over time and individual it is already not identified. We could want to estimate the average $\psi_j(i,t)$ though. If $\psi_j(i,t) = \psi_j(i)$ then there won't be any changes over time in worker machine allocation. We would never be able to disentangle the individual effect and the machine effect (no connected set). But if $\psi_j(i,t)$ is a function of both individual and time we would still probably have a problem with exogenous mobility.

Question 3

3.1) We want to estimate $E[Y_{it}(1) - Y_{it}(0)|D_{it} = 1]$. It is identified under the assumption that for $t \geq 2$ and all $g, g' \in \mathcal{G}$:

$$E[Y_{it}(0) - Y_{it-1}(0)|G_i = g] = E[Y_{it}(0) - Y_{it-1}(0)|G_i = g']$$

I.e. all groups have the same trend if they are not treated.

3.2) We know that the TWFE-regression is given by:

$$\tau_{TWFE} = E[\sum_{(g,t):D_{it}=1} w_{gt} \frac{N_{gt}}{N_1} \tau_{gt}]$$

With weights:

$$w_{gt} = \frac{\tilde{D}_{gt}}{\sum_{(g,t):D_{it}=1} \frac{N_{gt}}{N_1} \tilde{D}_{gt}}$$

$\tilde{D}$ is the residualized D_{it} regressed on time and group fixed effects. We get $\tilde{D}$:s:

$$\begin{aligned}
\tilde{D}_{22} &= D_{22} - \bar{D}_{g=2} - \bar{D}_{t=2} + \bar{D} = 1 - 1/2 - 1/2 + 1/2 = 1/2 \\
\tilde{D}_{23} &= 1 - 1/2 - 1 + 1/2 = 0 \\
\tilde{D}_{33} &= 1 - 1/2 - 1 + 1/2 = 0 \\
\tilde{D}_{34} &= 1 - 1/2 - 1/2 + 1/2 = 1/2
\end{aligned}$$

We get:

$$\sum_{(g,t):D_{it}=1} \frac{N_{gt}}{N_1} \tilde{D}_{gt} = 1/4 \times (1/2 + 1/2) = 1/4$$

Hence:

$$\begin{aligned}
w_{22} &= 2 \\
w_{23} &= 0 \\
w_{33} &= 0 \\
w_{34} &= 2
\end{aligned}$$

3.3) The switching estimator only compares movements $[0, 1]$ to $[0, 0]$ and $[1, 0]$ to $[1, 1]$. It wants to estimate ATE:s in cells (2,2) and (1,4), and leverages the above comparisons for this. For tihis to be valid we need the assumption for all $t \geq 2$ and all $g, g' \in \mathcal{G}$:

$$E[Y_{it}(1) - Y_{it-1}(1)|G_i = g] = E[Y_{it}(1) - Y_{it-1}(1)|G_i = g']$$

3.4) The identified group time ATE:s are:

$$\hat{\tau}_{22} = (4-1) - (3-2) = 2$$
$$\hat{\tau}_{24} = (3-4) - (3-5) = 1$$

Using the switching estimator we only use $\hat{\tau}_{22}$ and $\hat{\tau}_{14}$ which gives an $\hat{\tau}_{dCdH} = 1$ with equal weights.